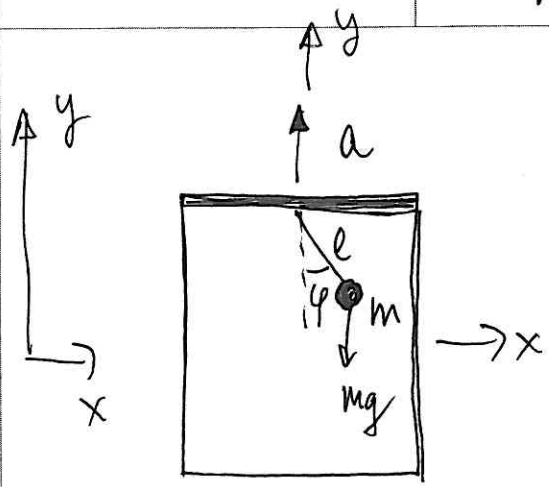
Midterm 1

P1



Pendulum in elevator:
support point moves as $y_0(t)$

$$\begin{cases} y = y_0(t) - l\cos\varphi \\ x = l\sin\varphi \end{cases}$$

$$L = \frac{1}{2} m \left[(l\cos\varphi\,\dot\varphi)^2 + (\dot y_0 + l\sin\varphi\,\dot\varphi)^2 \right] - mg(y_0 - l\cos\varphi)$$

$$= \frac{1}{2} m l^2 \dot\varphi^2 + \frac{1}{2} m \dot y_0^2 + m l\, \dot y_0 \sin\varphi\, \dot\varphi$$
$$- mg\, y_0 + mgl\cos\varphi$$

Lagrangians are defined upto time derivative of some function: L and $L + \frac{df}{dt}$ for arbitrary f result in the same equations of motion.

$\Rightarrow$ we can drop terms $\frac{1}{2} m \dot y_0^2 - mg\, y_0$ from L.

Also, $m l \sin\varphi\, \dot\varphi\, \dot y_0 = \frac{d}{dt}(-m l \cos\varphi\, \dot y_0) + m l \cos\varphi\, \ddot y_0$

$$\Rightarrow L = \frac{1}{2} m l^2 \dot\varphi^2 + mgl\cos\varphi + m l \cos\varphi\, \ddot y_0 + \underbrace{\frac{d}{dt}(-m l\, \dot y_0 \cos\varphi)}_{\text{full } t\text{-derivative, can be droped.}}$$

Note: 1 coordinate $= \varphi$
y_0 is <u>not</u> a coordinate.

Equations of motion: $\frac{d}{dt}\frac{\partial L}{\partial \dot\varphi} = \frac{\partial L}{\partial \varphi}$

$$m\ell^2 \ddot\varphi = m\ell(g+\ddot y_0)\frac{\partial}{\partial\varphi}\cos\varphi$$

$$\Rightarrow \quad \ddot\varphi = -\frac{g+\ddot y_0}{\ell}\sin\varphi \qquad \left(\Rightarrow V_{eff} = -m\ell(g+\ddot y_0)\cos\varphi\right)$$

Small oscillations $\Rightarrow \sin\varphi \approx \varphi$ (expand near min at $\varphi=0$)

$$\ddot\varphi = -\omega^2\varphi \text{ , where } \omega = \sqrt{\frac{g+\ddot y_0}{\ell}}$$

Thus, if motion is such that $\ddot y_0 = 0$

(which includes uniform motion $y_0 = v_0 t$)

then $\omega = \sqrt{\frac{g}{\ell}}$ = standard frequency.

If $\ddot y_0 = a$ (motion with constant acceleration)

than $\omega = \sqrt{\frac{g+a}{\ell}}$

Note that if acceleration is downward $(a<0)$

then $\omega = \sqrt{\frac{g-|a|}{\ell}}$. If $|a|>g$, then $\varphi=0$

is not a minimum anymore, but $\varphi=\pi$ is!

oscillations frequency then becomes

$\omega' = \sqrt{\frac{|a|-g}{\ell}}$ for $|a|>g$.

(P2) Central potential $U(r)$ such that

(a) $$F(r) = -\frac{dU(r)}{dr} = -\frac{K}{r^2} e^{-r/a}$$

$$L = \frac{1}{2} m\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) - U(r)$$

$$\frac{\partial L}{\partial \varphi} = 0 \Rightarrow \frac{\partial L}{\partial \dot{\varphi}} = mr^2\dot{\varphi} = M_z \text{ is conserved.}$$

$$-\frac{dU(r)}{dr} = F(r) = -\frac{K}{r^2} e^{-r/a}$$

$$\Rightarrow m\ddot{r} = \frac{\partial L}{\partial r} = mr\dot{\varphi}^2 - \frac{dU(r)}{dr} = mr\left(\frac{M_z}{mr^2}\right)^2 - \frac{dU}{dr}$$

$$m\ddot{r} = -\frac{\partial}{\partial r}\left(\frac{M_z^2}{2mr^2} + U(r)\right) = -\frac{\partial}{\partial r} U_{eff}(r) \quad (0)$$

Effective potential $$U_{eff}(r) = \frac{M_z^2}{2mr^2} + U(r)$$

(b) Find the minimum of effective potential

$$\frac{\partial}{\partial r} U_{eff}(r)\Big|_{r=R} = 0$$

$$-\frac{M^2}{mr^3} + \frac{\partial U}{\partial r}\Big|_R = 0 \quad \text{but } \frac{\partial U}{\partial r} = \frac{K}{r^2} e^{-r/a}$$

$$\Rightarrow \frac{K}{R^2} e^{-R/a} = \frac{M^2}{mR^3} \quad (1)$$

$$KRe^{-R/a} = \frac{M^2}{m}.$$

Check that it is minimum:

$$\left.\frac{d^2 U_{eff}}{dr^2}\right|_{r=R} = \frac{d}{dr}\left(-\frac{M^2}{mr^3} + \frac{K}{r^2}e^{-r/a}\right)_{r=R} =$$

$$= \left(\frac{3M^2}{mr^4} - \frac{2K}{r^3}e^{-r/a} - \frac{K}{ar^2}e^{-r/a}\right)_{r=R}$$

$$\Rightarrow \left.\frac{d^2U}{dr^2}\right|_{r=R} = \frac{3M^2}{mR^4} - \frac{2Ke^{-R/a}}{R^3} - \frac{K}{aR^2}e^{-R/a} =$$

(Use eq. 1) $$= \frac{1}{R^4}\left(\frac{3M^2}{m} - 2KRe^{-R/a} - \frac{KR^2}{a}e^{-R/a}\right)$$

$$= \frac{1}{R^4}\left(\frac{3M^2}{m} - 2\frac{M^2}{m} - \frac{R}{a}\frac{M^2}{m}\right)$$

$$= \frac{1}{R^4}\frac{M^2}{m}\left(1 - \frac{R}{a}\right)$$

Need to have $\left.\frac{d^2U}{dr^2}\right|_R > 0 \Rightarrow \frac{M^2}{mR^4}\left(1-\frac{R}{a}\right) > 0$

(c) $M = mR^2\omega = mRv$

$\Rightarrow Re^{-R/a} = \frac{(mRv)^2}{mK} \Rightarrow v = \frac{K}{mR}e^{-R/a}$

$\Rightarrow \boxed{R < a}$ $\frac{R}{a} < 1$

(d) Back to eq. (0): $r = R + \eta$

$$\Rightarrow m\ddot{\eta} = -\frac{d}{dr}U_{eff}(R+\eta) = -\frac{dU_{eff}(R)}{dr} - \frac{d^2U_{eff}(R)}{dr^2}\eta$$

$$\hookrightarrow U_{eff}(R) + \frac{dU_{eff}(R)}{dr}\eta$$

$$\Rightarrow \ddot{\eta} = -\frac{M^2}{m^2R^4}\left(1-\frac{R}{a}\right)\eta \Rightarrow \omega = \frac{M}{mR^2}\sqrt{1-\frac{R}{a}}$$

(P3) (a) $\tau = \int_0^t \sqrt{1-\frac{v_0^2}{c^2}}\,dt = \sqrt{1-\frac{v_0^2}{c^2}}\;t$

$$\Rightarrow \sqrt{1-\frac{v_0^2}{c^2}} = \frac{3}{5} \Rightarrow \frac{v_0^2}{c^2} = \frac{16}{25} \Rightarrow v_0 = \frac{4}{5}c.$$

(b) $v(t) = \frac{4c}{5}\sqrt{1-\frac{t}{T}}$

$$3 = \tau = \int_0^t \sqrt{1-\frac{16}{25}\left(1-\frac{t}{T}\right)}\,dt = \int_0^{t/T} \sqrt{\frac{9}{25}+\frac{16t}{25T}}\,d\left(\frac{t}{T}\right)\cdot T$$

$$\tau = \frac{T}{5}\int_0^{t/T} \sqrt{9+16x}\,dx = \frac{T}{5}\,\frac{(9+16x)^{3/2}}{24}\Big|_0^{t/T}$$

$$3 = \tau = \frac{T}{5}\left[\frac{\left(9+16\frac{t}{T}\right)^{3/2}}{24} - \frac{9^{3/2}}{24}\right]$$

$$3.24 = \left(9+\frac{16}{5}t\right)^{3/2} - 27$$

$$\left(9+\frac{16}{5}t\right)^{3/2} = 72+27 = 99$$

$$\frac{16}{5}t = (99)^{2/3} - 9, \quad t = \frac{5}{16}\left[(99)^{2/3} - 9\right] = 3.875 \text{ years}$$

$= 3$ year 319 days.